

班级: _____

姓名: _____

编号: _____

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2.4 解:

$$y[n] = x[n] * h[n]$$

$$= \begin{cases} 0, & n \leq 6 \\ n-6, & 7 \leq n \leq 12 \\ 6, & 13 \leq n \leq 18 \\ 24-n, & 19 \leq n \leq 23 \\ 0, & n \geq 24 \end{cases}$$

$$\text{故 } y(t) = C \cdot e^{-4t} + \frac{1}{3+3j} [e^{(-1+3j)t} - e^{-4t}]$$

由 $y(t) = 0$, 得 $C = 0$,

$$\text{故 } y(t) = \frac{1}{3+3j} [e^{(-1+3j)t} - e^{-4t}]$$

~~设特解为 $C_1(t)e^{-4t}$~~

2.11 解: (a)

$$y(t) = \begin{cases} 0, & t \leq 3 \\ \frac{1}{3}(1 - e^{-3t+9}), & 3 < t \leq 5 \\ \frac{1}{3}(e^{-3t+15} - e^{-3t+9}), & t > 5 \end{cases}$$

~~则 $C_1(t) =$~~

$$(b) \text{ 令 } z(t) = e^{-t} \cdot e^{3tj} \cdot u(t) = e^{(-1+3j)t} \cdot u(t)$$

$$\text{即 } \operatorname{Re}\{z(t)\} = x(t)$$

$$(b) g(t) = [\delta(t-3) - \delta(t-5)] * h(t)$$

$$= \begin{cases} 0, & t \leq 3 \\ e^{-3t+9}, & 3 < t \leq 5 \\ e^{-3t+9} - e^{-3t+15}, & t > 5 \end{cases}$$

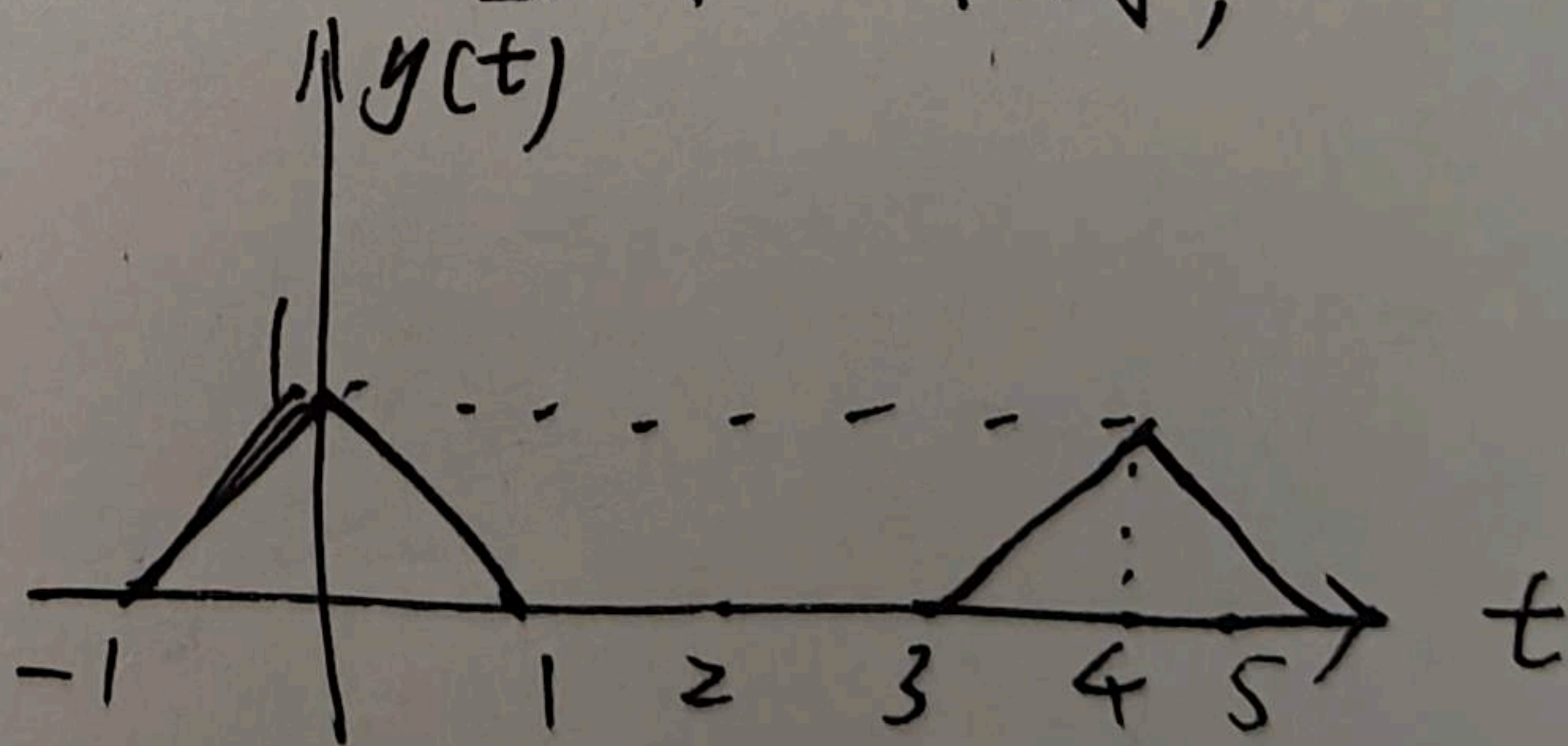
$$\text{故 } \frac{d}{dt} \operatorname{Re}\{y(t)\} + 4 \operatorname{Re}\{y(t)\}$$

$$= \frac{d}{dt} \operatorname{Re}\{z(t)\} = x(t)$$

故输入为 $x(t)$ 时输出即

$$\operatorname{Re}\{y(t)\} = \frac{1}{6} e^{-t} (\cos 3t - e^{3t} + \sin 3t)$$

$$2.23 \text{ 解: } y(t) = \sum_{k=-\infty}^{+\infty} h(t-kT)$$

(a) 当 $T = 4$ 时,

17 解: (a) 求齐次解:

$$r + 4 = 0$$

$$r = -4$$

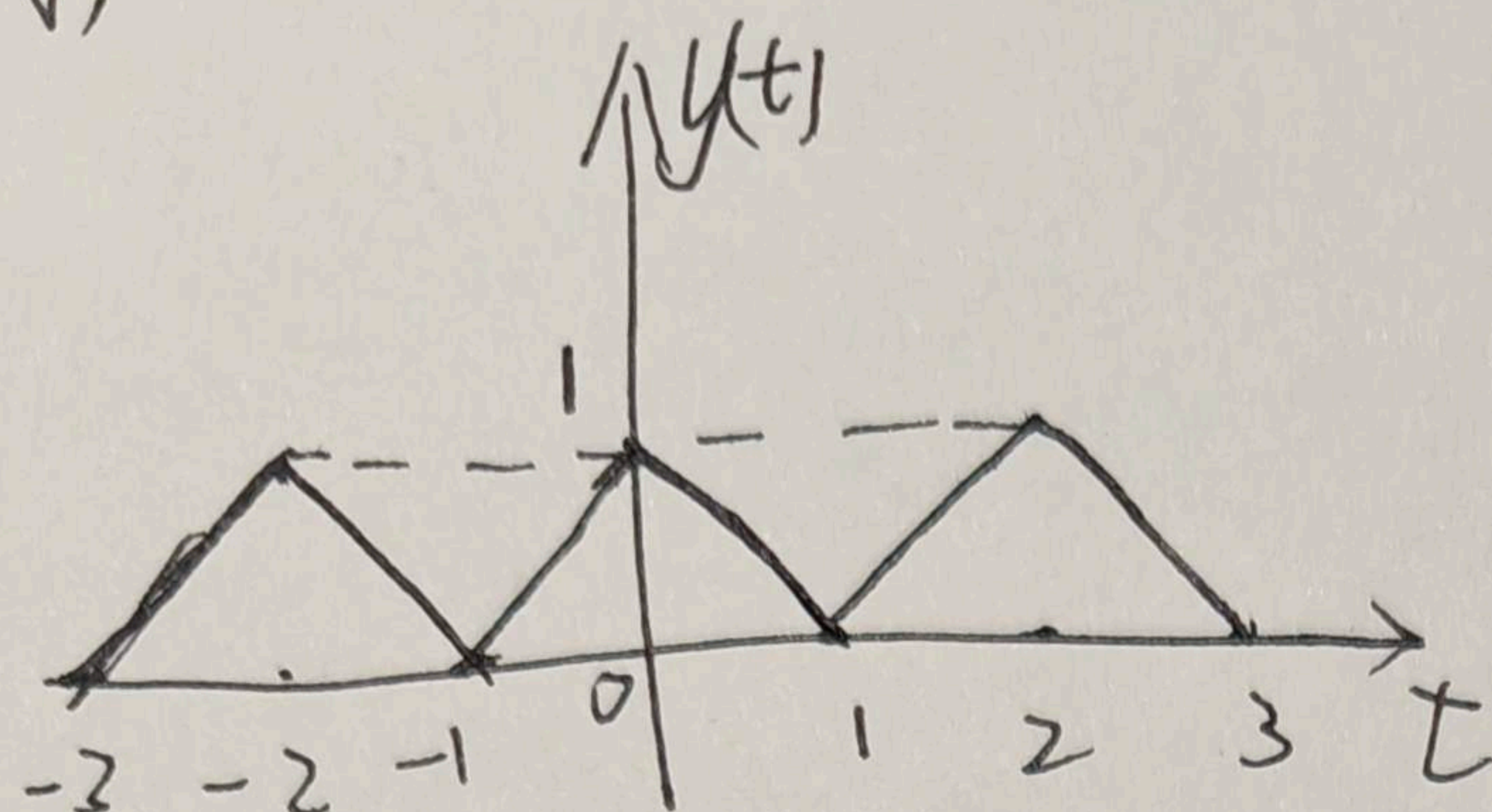
$$y_h(t) = C e^{-4t}$$

~~由 $y(t)$~~ 设特解为 $C(t)e^{-4t}$,

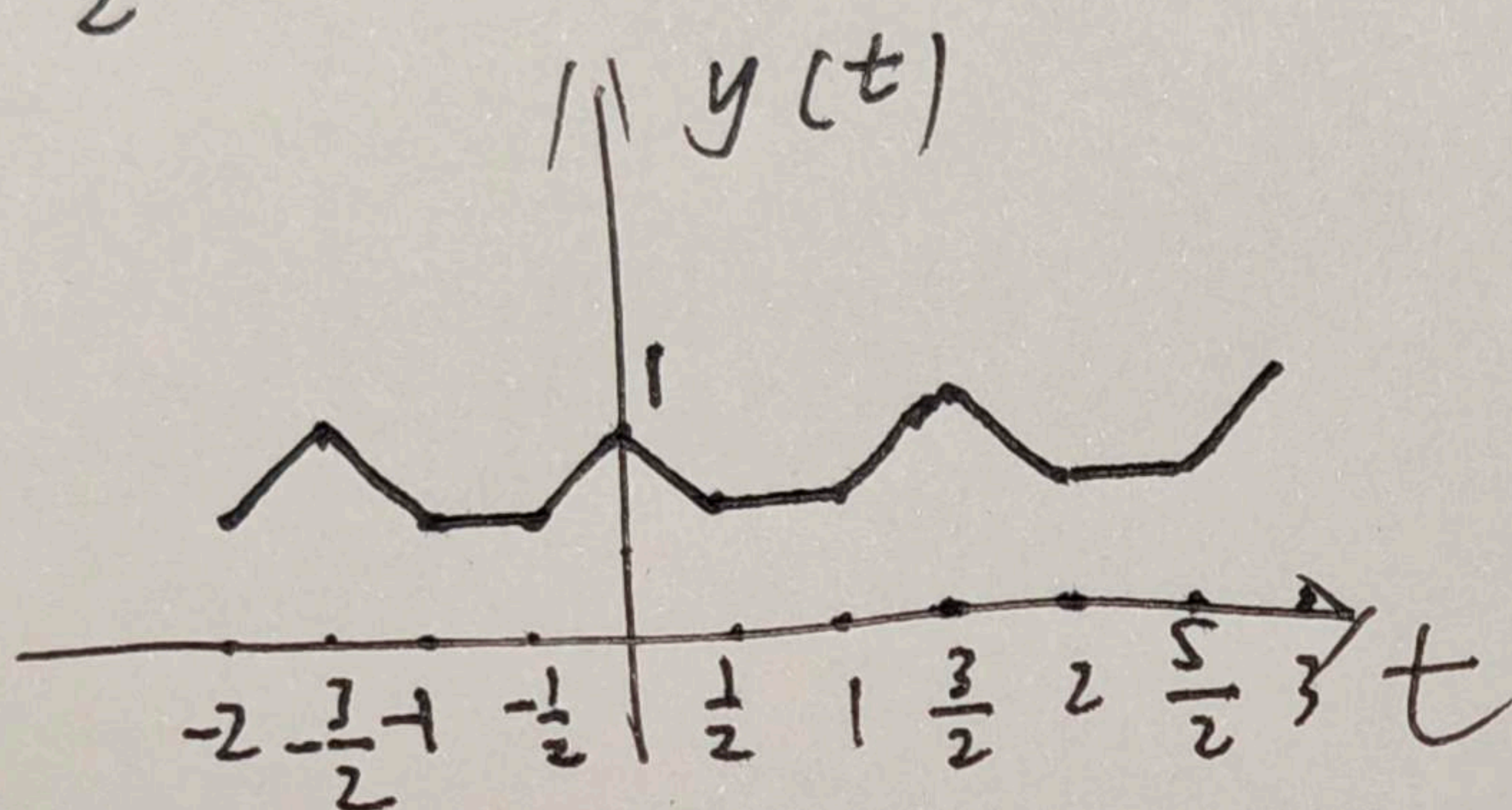
$$\text{则: } C'(t)e^{-4t} - 4C(t)e^{-4t} + 4C(t)e^{-4t} = e^{(-1+3j)t} u(t)$$

$$C(t) = \frac{1}{3+3j} [e^{(3+3j)t} - 1]$$

(2) $T=2$ 时,



(3) $T=\frac{3}{2}$ 时,



(4) $T=1$ 时,

